American Computer Science League

2023 Finals ● Short Problems ● Junior Division

1. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{AB} + \overline{A(B + C)}$$

A. $A + \overline{B}$
B. $\overline{A} + B$
C. $A + B$
D. $\overline{A} + \overline{B}$
E. None of the above

2. Boolean Algebra

How many ordered triples make the following Boolean expression FALSE?

$$A\overline{B} + A(\overline{B} + C) + \overline{B}(\overline{A} + \overline{C})$$

A. 2
B. 3
C. 4
D. 5
E. None of the above

3. Bit-String Flicking

Evaluate the following expression:

010101 OR 101101 AND NOT 011011

A. 111101
B. 100100
C. 110101
D. 101010
E. None of the above

4. Bit-String Flicking

Evaluate the following expression:

(LCIRC-2 (RSHIFT-1 (NOT (LSHIFT-1 (RCIRC-2 11010)))))

A. 00101
B. 10011
C. 01100
D. 10110
E. None of the above

5. Recursive Functions

Find $f(f(f(f(8))))$ given:

$$f(x) = \begin{cases} 2 & \text{if } x \leq 1 \\ 2 \cdot f(x-1) - x & \text{if } x > 1 \end{cases}$$

A. -2
B. 2
C. 1
D. -9
E. None of the above

6. Recursive Functions

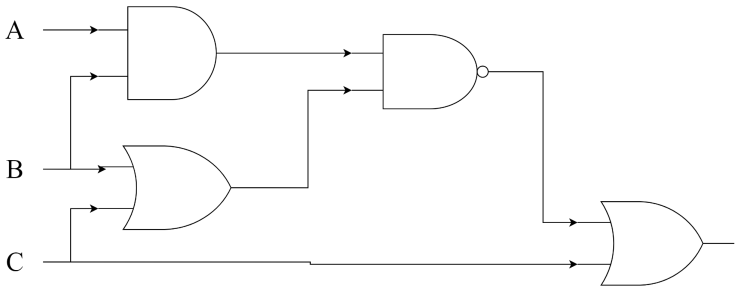
Find $f(21)$ given:

$$f(x) = \begin{cases} 2 + f(x-3) & if\ x \geq 12 \\ f(x-2) - 1 & if\ 5 < x < 12 \\ x+2 & if\ x \leq 5 \end{cases}$$

A. 6
B. 7
C. 11
D. 13
E. None of the above

7. Digital Electronics

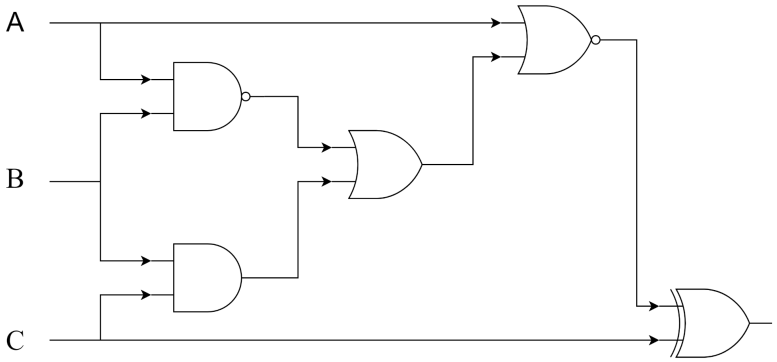
Simplify the Boolean expression for the following circuit if the value of B is always 1.



A. $A + C$
B. $\overline{A} + C$
C. $A + \overline{C}$
D. $\overline{A} + \overline{C}$
E. None of the above

8. Digital Electronics

How many ordered triples make the following digital circuit TRUE?



A. 0
B. 2
C. 4
D. 6
E. None of the above

9. Prefix-Infix-Postfix

Translate this prefix expression into postfix if all numbers are single digits.

+ + 2 / * 3 − 4 1 3 ^ 2 3

A. 2341−*3+/23^+
B. 2341−*3/+23^+
C. 2341*−3/+23^+
D. 2341−*3/+23+^
E. None of the above

10. Prefix-Infix-Postfix

Define: a # b = (a-b)(a+b)
Evaluate this postfix expression if all numbers are single digits:

3 2 # 5 4 # * 3 4 * − 4 2 # 1 − /

A. 3
B. 4
C. 5
D. 6
E. None of the above

11. Computer Number System

Convert 5272023_8 to hexadecimal.

A. 157323
B. 157213
C. 157513
D. 157413
E. None of the above

12. Computer Number Systems

Evaluate the following and express the result in hexadecimal:

$7_{10} + 77_8 + 777_{16}$

A. 7BD
B. 7B5
C. 7DB
D. 7CD
E. None of the above

13. Data Structures

Build a binary search tree for: HIPPOPOTAMUS
List the nodes from left to right at depth 3.

A. P
B. OSU
C. PT
D. OP
E. None of the above

14. Data Structures

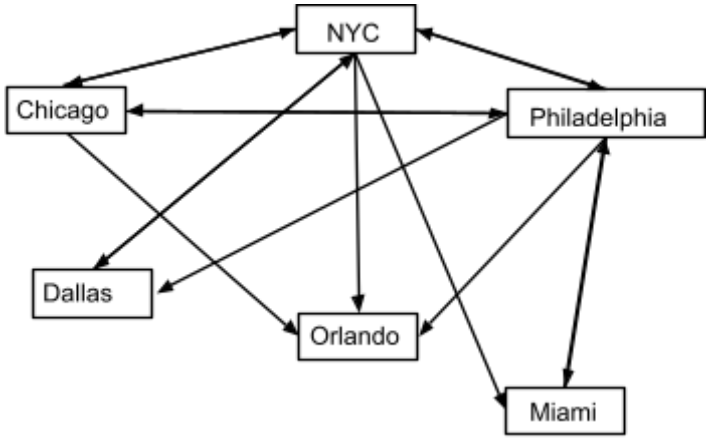
Given an initially empty queue and the following commands on the queue, read left to right first, what is the maximum number of items in the queue at any time?

PUSH(“P”),	PUSH(“R”),	X=POP(),	PUSH(“O”),
PUSH(“G”),	X=POP(),	PUSH(“R”),	PUSH(“A”),
X=POP(),	X=POP(),	PUSH(“M”),	X=POP(),
PUSH(“M”),	X=POP(),	PUSH(“E”),	X=POP(),
PUSH(“R”)			

A. 1
B. 2
C. 3
D. 4
E. None of the above

15. Graph Theory

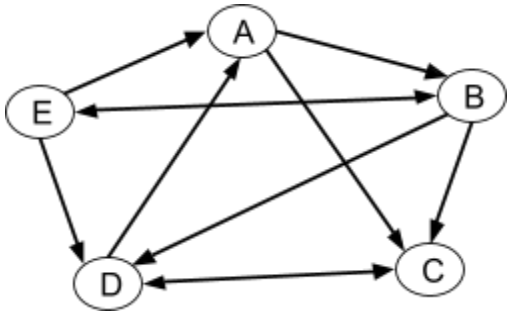
Consider the following graph showing flights between cities. Starting from NYC, find all possible routes to each of the other cities shown below, with zero or more stops, but without visiting any city more than once. However, if Miami is shut down due to a hurricane, how many fewer routes from NYC are there, again without visiting any city more than once?



A. 5
B. 6
C. 7
D. 8
E. None of the above

16. Graph Theory

How many pairs of vertices do not have a path of length 2? Include starting and ending at the same vertex.



A. 8
B. 13
C. 16
D. 22
E. None of the above

17. What Does This Program Do?

What is the output for the following program?

```
a = 30: b = 40: c = 50: d = 16
if a + b > c then
     d = d / 2
else
     d = d - 1
end if
if c - b < a then
     d = d % 3
end if
d = d + 1
if d % 2 != 0 && d % 3 == 0 then
     output (b + c) / a
else
     output c / (b - a)
end if
```

A. 3
B. 5
C. -5
D. 6
E. None of the above

18. What Does This Program Do?

What is the output when the following program is executed?

```
s = 0 : p = 0
n = 84
while n >= 2
     f = 2
     while n % f != 0
          f = f + 1
     end while
     if f != p then
          p = f
          s = s + f
     end if
     n = n / f
end while
output s
```

A. 42
B. 23
C. 14
D. 12
E. None of the above

19. What Does This Program Do?

Given that array **arr** contains the 10 values below, what is the final value of **count**?

8	6	2	9	3	5	7	1	4	0

```
count = 0
for x = 0 to 8
     for y = x+1 to 9
          if x == arr(y) then
               z1 = x: z2 = y
               while z1 < z2
                    t = arr(z1)
                    arr(z1) = arr(z2)
                    arr(z2) = t
                    count = count + 1
                    z1 = z1 + 1
                    z2 = z2 - 1
               end while
          end if
     next y
next x
```

A. 14
B. 16
C. 17
D. 18
E. None of the above

20. What Does This Program Do?

What is the output for the following program? Remember that val("9") = 9 and str(9) = "9".

```
ans = ""
s = "101010100111"
for x = len(s)-1 to 0 step -3
     p = 1
     v = 0
     for y = 0 to 2
          v = v + p * val(s[x-y])
          p = p * 2
     next y
     ans = str(v) + ans
next x
output ans
```

A. 54314
B. 5247
C. 1127
D. 74362
E. None of the above